## Resistivity (ρ)

Let us consider a conductivity cell in which the electrolytic solution is confined between the two electrodes having cross sectional area (A) and are separated by a distance '$l$' . Like the metallic conductor, the resistance of such an electrolytic solution is also directly proportional to the length ($l$) and inversely proportional to the cross sectional area (A).

$$R \alpha \frac{l}{A}$$

$$R = \rho \frac{l}{A} \qquad (9.2)$$

Where ρ (rho) is called the specific resistance or resistivity, which depends on the nature of the electrolyte.

If $\frac{l}{A} = 1\ m^{-1}$, then, $\rho = R$ . Hence the resistivity is defind as the resistance of an electrolyte confined between two electrodes having unit cross sectional area and are separated by a unit distance. The ratio $\left(\frac{l}{A}\right)$ is called the cell constant, Unit of resistivity is ohm metre (Ωm).

## Conductivity

It is more convenient to use conductance rather than resistance. The reciprocal of the resistance $\left(1/R\right)$ gives the conductance of an electrolytic solution. The SI unit of conductance is Siemen (S) .

$$C = \frac{1}{R} \qquad \text{..... (9.3)}$$

Substitute (R) from (9.2) in (9.3)

$$\Rightarrow \text{ i.e., } C = \frac{1}{\rho}.\frac{A}{l} \qquad \text{.....(9.4)}$$

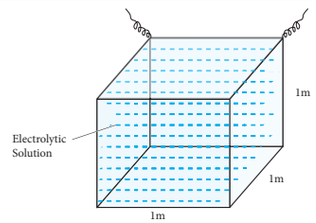



**Fig 9.2 conductivity of a cube of an electrolytic solution**

The reciprocal of the specific resistance $\left(1/\rho\right)$ is called the specific conductance (or) conductivity. It is represented by the symbol κappa (κ).

Substitute $\frac{1}{\rho} = \kappa$ in equation (9.4) and rearranging

$$\Rightarrow \kappa = C.\left(\frac{l}{A}\right) \qquad \text{.....(9.5)}$$

Unit of κ

$$\kappa = \frac{1}{R} \cdot \frac{l}{A} \left(\frac{1}{\text{ohm}} \cdot \frac{m}{m^2}\right)$$

$$= \text{ohm}^{-1}\ m^{-1} = \text{mho}\ m^{-1} \text{ (or) } Sm^{-1}$$

If $A = 1m^2$ and $l = 1m$ ; then $\kappa = C$.

The specific conductance is defined as the conductance of a cube of an electrolytic solution of unit dimensions(Fig 9.2). The SI unit of specific conductance is $Sm^{-1}$ .

## Example

A conductivity cell has two platinum electrodes separated by a distance 1.5 cm and the cross sectional area of each electrode is 4.5 sq cm. Using this cell, the resistance of 0.5 N



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